25900564 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEPHY207 - PROPERTIES OF MATTER

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. If a cylinder with radius (R) and Length (L) is given, give the expression to find out the twisting couple of the cylinder [A] / [CO2]
2. Define the terms plane of bending and axis of bending. [U] / [CO1, CO2]
3. State the relation connecting Young's modulus, Bulk modulus and Poisson's ratio. [U] / [CO1, CO2]
4. How can we define piezoelectricity? [K] / [CO2]
5. What you mean by a cantilever? [U] / [CO1, CO2]
6. State the assumptions made in deriving Bernoulli’s principle. [U] / [CO3]
7. What is meant by excess of pressure? What is its SI unit? [U] / [CO3]
8. How does the viscosity of a fluid affect its terminal velocity? [U] / [CO3]
9. What is meant by surface energy? [K] / [CO3]
10. Define time period of a wave. [U] / [CO4]

11. State any four properties of ultrasonic waves. [U] / [CO6]

12. Write the differential equation for a one dimensional wave travelling along the positive x direction. [U] / [CO4]

13. What are beats? [U] / [CO5]

14. Define a longitudinal wave. Give two examples. [U] / [CO5]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15. Explain about the different elastic behavior of solids. [U] / [CO1]

16. How can we compare moments of inertia using torsion pendulum? [K] / [CO2]

17. Draw and explain the stress strain diagram. [An] / [CO1, CO2]

18. Explain the molecular theory of surface tension. [U] / [CO3]

19. Calculate the Reynold's number for water flowing through a pipe of diameter 0.1 m at avelocity 2 m/s. Given density of water=1000 kg/m^3 and viscosity of water=0.001 Pa.s [A] / [CO3]

20. How is ultrasonic is used in flaw detection? [U] / [CO6]

21. Calculate the velocity with which a source of frequency 10000 Hz should approach an observer at rest in order to produce a Doppler shift of 200 Hz. Velocity of sound 332m/s. [U] / [CO6]

22. Explain the significance of Mach number in classifying the motion of an aircraft. [An] / [CO5]

[5x6 = 30]